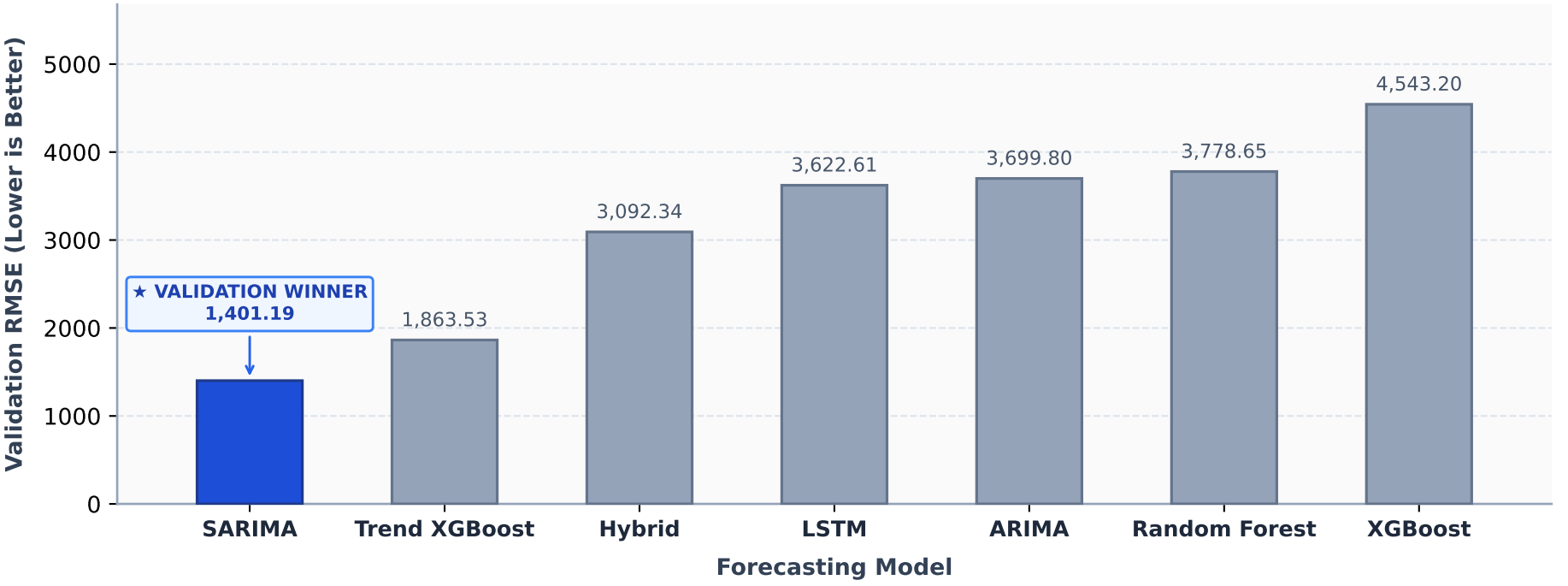
# Electricity Demand — Validation-Driven Model Benchmark

Model selection based on validation performance

**Selected on Validation Set | Criterion: Minimum Validation RMSE**

## Validation RMSE by Candidate Model




| Rank | Model | Val RMSE ↓ | Val MAE ↓ | Val MAPE ↓ | Val R² ↑ | Test RMSE ↓ |
|---|---|---|---|---|---|---|
| **#1** | **★ SARIMA (Recommended Winner)** | **1401.19** | 952.43 | 0.50% | 0.9951 | 1585.32 |
| #2 | Trend XGBoost | **1863.53** | 1334.73 | 0.71% | 0.9914 | 2552.53 |
| #3 | Hybrid | **3092.34** | 2304.61 | 1.21% | 0.9763 | 4096.27 |
| #4 | LSTM | **3622.61** | 2821.83 | 1.49% | 0.9675 | 4861.41 |
| #5 | ARIMA | **3699.80** | 2765.71 | 1.46% | 0.9661 | 4816.97 |
| #6 | Random Forest | **3778.65** | 2662.97 | 1.38% | 0.9647 | 4998.18 |
| #7 | XGBoost | **4543.20** | 2867.96 | 1.45% | 0.9489 | 6330.94 |

*Model selection is performed using validation RMSE. The held-out test set is reserved for final unbiased evaluation.*

**Selected Model: SARIMA (Validation Winner) | Validation RMSE: 1401.19**